



A computerized method for loaded tooth contact analysis of high-contact-ratio spur gears with or without flank modification considering tip corner contact and shaft misalignment



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ABSTRACT

High-contact-ratio (HCR) spur gears have the advantages of reducing the acting loads on the teeth, as well as noise and vibration. However, the phenomenon of tip corner contact and the influence of shaft misalignments can cause unexpected damage and vibration. Flank modification is a solution for these influences. An efficient computerized tool for loaded tooth contact analysis (LTCA) is proposed in this paper that can be used to design HCR gears, while taking into consideration the conditions of tip corner contact and shaft misalignment. The LTCA method is developed based on the influence coefficient method and considers both the deformation of the loaded teeth and the twist deformation of the shaft. The variations of the shared loads and contact stress of a HCR spur gear pair with lead crowning and relieved profiles are analyzed considering the influence of the amount of shaft misalignment and relief. The contact patterns as well as the stress distribution on the engaged flanks in various cases are also simulated.

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1. Introduction

Increased demand for less vibration and noise in gear transmission has led to a focusing of attention on the development of different design concepts. One strategy is to design gear drives with a high contact ratio (HCR), i.e., greater than 2. Although the loading characteristics of spur gears are poorer than those of helical gears, they are the simplest among the cylindrical gears to design, so they are still applied widely for various power transmission tasks, e.g., in planetary gear drives. It might also be meaningful to transfer the research results obtained for some critical issues related to spur gears to helical gears, especially related to the interference contact of tooth pairs due to tip corner contact or shaft misalignment which will not only cause rapid damage to the flanks due to concentrated stress and unfavorable tooth lubrication [1], but also vibration and noise due to high-frequency harmonics of the transmission error. In order to avoid the occurrences of these problems which are often found in spur gear drives with a high contact ratio, flank modification is applied. Developing a loaded tooth contact analysis (LTCA) method to explore the influence of the flank modification on the contact characteristics in such a gear drive is hence essential, especially for consideration of the tip corner contact and the shaft misalignment.

A high contact ratio, i.e., greater than 2, in a spur gear drive means that the number of contact tooth pairs during meshing will be at least either 2 or 3. The variation of the shared load and the contact stress during the gear mesh is quite different from those of the normal spur gears (where the contact ratio is less than 2). In the past few years, many research studies have been carried out analyzing the contact characteristics of multiple contact tooth pairs. The methods can be divided into three categories: analytical, numerical and finite element methods. For example, Elkholy developed a stiffness approach for the calculation of shared loads in HCR gears based on analytical equations for the deformation of the loaded teeth [2]. Pedrero et al. [3] proposed a load distribution model based on the minimum

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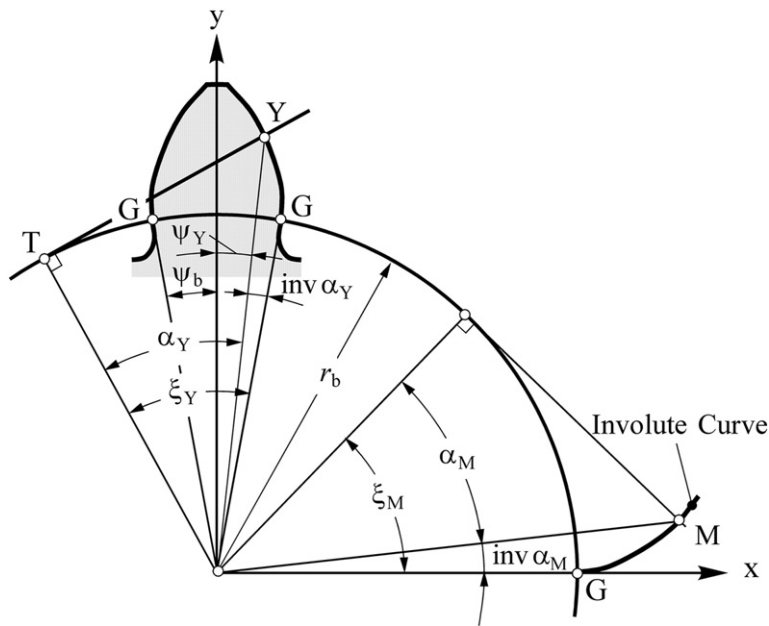


Fig. 1. Definition of unmodified tooth flank.

elastic potential criteria for the calculation of tooth bending strength and surface durability. Based on this model, Sánchez et al. [4] further developed a method for the calculation of the contact stress along the line of action. However, none of the methods mentioned above can deal with the case of flank modification. Apart from the application of finite element method (FEM) for the calculation of the tooth bending and contact stress of HCR spur gears [5,6], Li combined FEM with calculation programs to develop a face-contact model and a 3D geometry model of engaged teeth with which to conduct LTCA for the spur gears with different addendums and contact ratios [7]. Some calculation programs based on various numerical methods for load distribution have also been developed, e.g. LVR at the TU Dresden (Germany) and LDP at the Ohio State University (USA). Wu and Tsai proposed a computerized LTCA approach based on

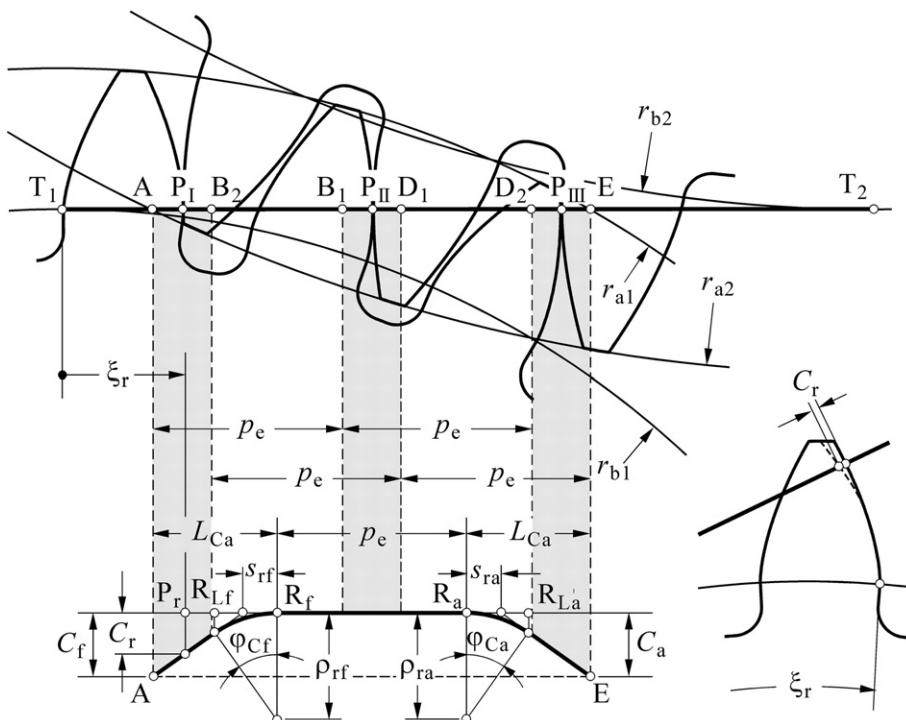


Fig. 2. Definition of linear tip and root relief with smoothing.

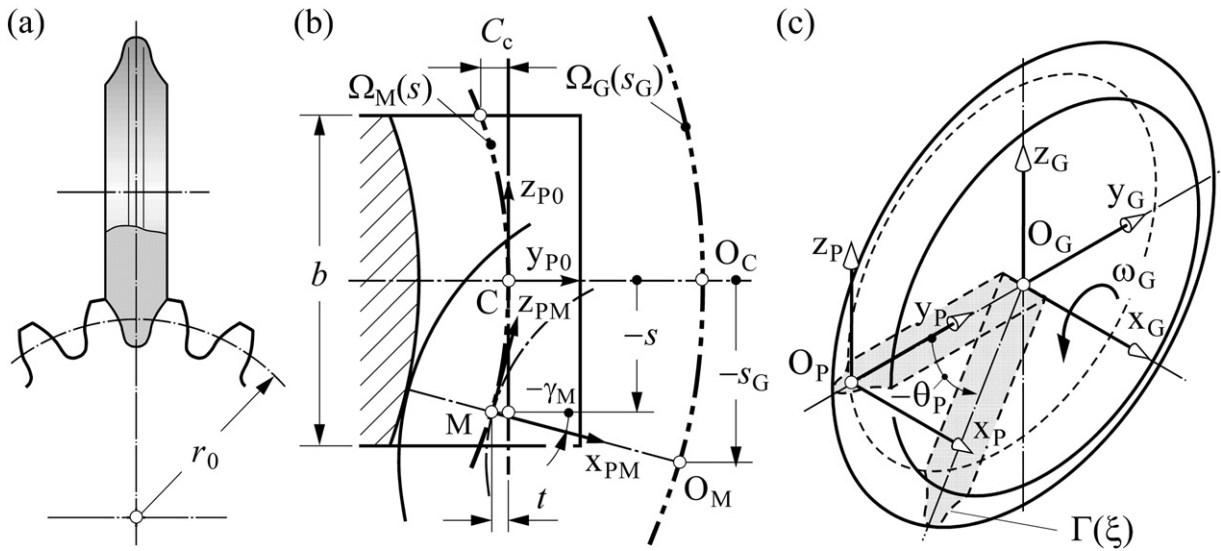


Fig. 3. Geometric relation for lead crowning.

the influence coefficient method to calculate the contact stress of multiple gear tooth pairs numerically. The contact pattern and stress distribution on the flanks, and even the non-Hertzian contact can be also simulated [8,9].

In addition to the multiple tooth-pair contact, the phenomenon of interference contact of the loaded teeth on the tip corner outside the normal line of action can also not be ignored. However, there have been few studies about the problem of tip corner contact. Munro et al. did develop an analytical approach for calculating the transmission errors of HCR spur gears with corner contact [10] and also conducted an experiment for measuring the tooth pair stiffness to validate the theoretical results [11]. Based on two simplified contact model, Sato and Umezawa [12] developed a calculation method and also conducted an experiment to obtain the mesh stiffness with tip corner contact. Yeh et al. [13] used the LTCA method [8] to analyze the contact characteristics of spur gears with the normal contact ratio.

The influence of shaft misalignment on the contact characteristics, another important issue for HCR gears, is often analyzed together with lead modification. In addition to the often applied FEM [14,15], a computerized approach would help to make

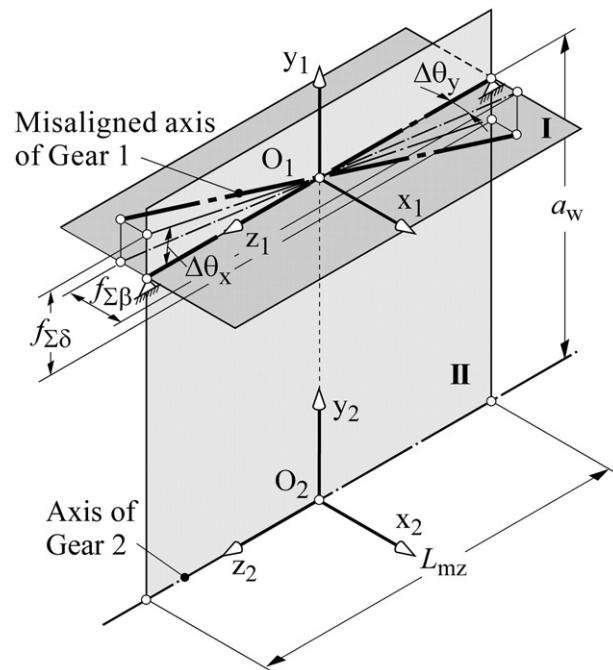


Fig. 4. Deviations of the misaligned shaft.

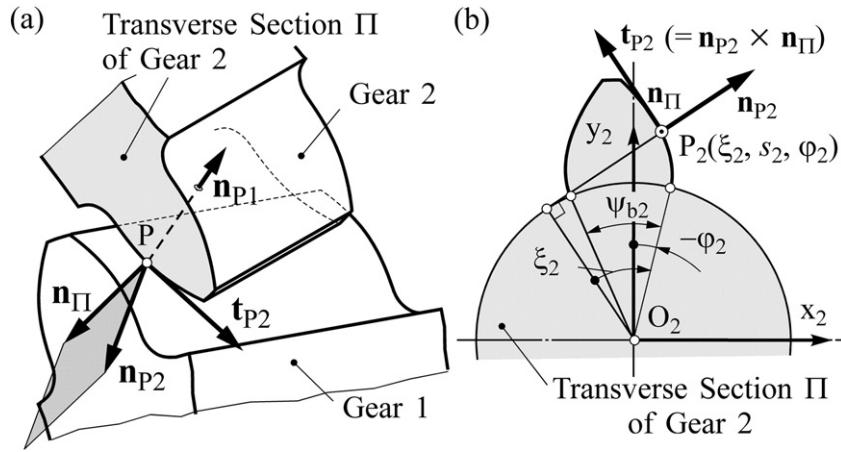


Fig. 5. Geometric relation for face-end edge contact.

analysis more efficient. Houser discussed the sources of misalignment and analyzed the contact characteristics of perfect and modified involute gears by using the developed computer program LDP [16]. Hotait and Kahraman [17,18] studied the influence of shaft misalignment on load distribution and tooth bending stress experimentally with different amounts of lead crowning, as well as compared the results with those obtained from LDP. Artoni, Kahraman et al. [19] used the program LDP to optimize the flank modification of gears within a range of applied torque and misalignment. Profile relief is usually applied to reduce the influence of tip corner contact. Many studies related to the design and analysis of this method can also be found. For example, Lin [20,21] analyzed the dynamic behaviors of spur gears with linear and parabolic tip relief. Beghini et al. [22] proposed a method for smoothing tip relief curve to obtain smooth variation of contact stress and minimum transmission errors.

The aim of this current paper is to develop a computerized LTCA approach to explore the contact characteristics of HCR gears with and without flank modification in the presence of shaft misalignment. The influence of tooth contact on the tip corner and the face-end, which has rarely been discussed in the literatures, is also considered and discussed here. The variation in shared loads and contact stresses, the loaded transmission errors, as well as the contact patterns with distributed stresses will be calculated and simulated.

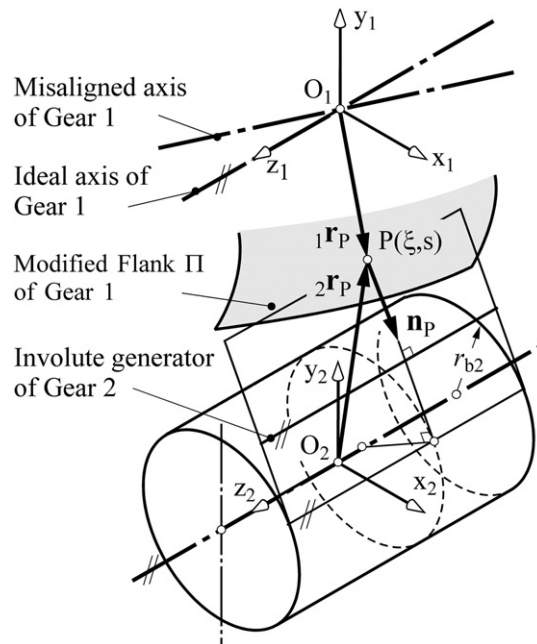


Fig. 6. Relation of shaft misalignment and lead crowning.

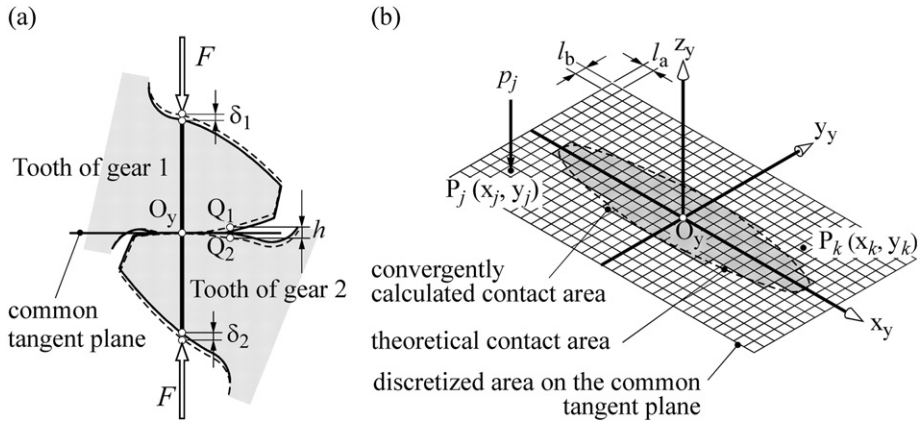


Fig. 7. (a) Relation of deformation and load (b) meshing on the common tangent plane.

2. Mathematical model of tooth surfaces

2.1. Unmodified tooth flank

The unmodified tooth flank is considered to be an involute curve and thus can be generated based on the equation for the involute curve, as shown in Fig. 1. The involute curve generated from the origin point G and the coordinates of the point M(x_M, y_M) on the curve is

$$\begin{aligned} x_M &= r_b \cdot (\cos \xi_M + \text{arc} \xi_M \cdot \sin \xi_M) \\ y_M &= r_b \cdot (\sin \xi_M - \text{arc} \xi_M \cdot \cos \xi_M) \end{aligned} \quad (1)$$

The coordinates of the point Y(x_Y, y_Y) on the tooth flank which is symmetrical to the y axis can be derived by transforming the coordinates of the involute curve from Eq. (1) as

$$\begin{aligned} x_{YL,R} &= \mp x_M \cdot \sin \psi_b \pm y_M \cdot \cos \psi_b \\ y_{YL,R} &= +x_M \cdot \cos \psi_b + y_M \cdot \sin \psi_b \end{aligned} \quad (2a)$$

where the parameter ψ_b is the tooth thickness half angle according to DIN 3960 [23],

$$\psi_b = \psi + \text{inv} \alpha_p = \frac{\pi + 4 \cdot x \cdot \tan \alpha_n}{2 \cdot z} + \text{inv} \alpha_p. \quad (2b)$$

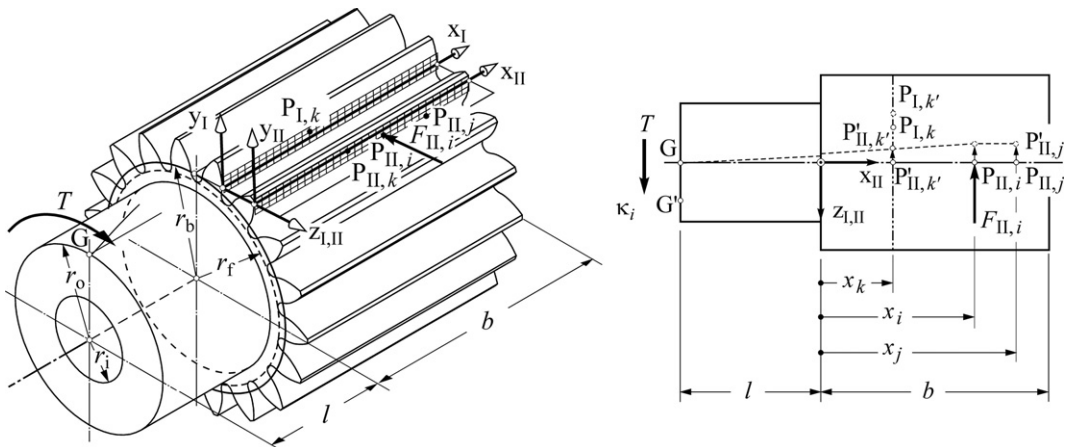


Fig. 8. Twist of the gear shaft.

Table 1

Gearing data for the analyzed spur gear pair.

Items	Unit	Pinion	Gear
Center distance a_w	mm	88	
Number of teeth z	–	27	44
Normal module m_n	mm	2.5	
Pressure angle α_p	deg	20	20
Normal profile shifting coefficient x	–	–0.0534	–0.2368
Actual profile shifting coefficient x_{act}	–	–0.1138	–0.2972
Face width b	mm	45	42
Addendum coefficient h_{ap}^*	–	1.25	1.25
Dedendum coefficient h_{dp}^*	–	1.4	1.4
Root radius factor ρ_p^*	–	0.39	0.39
Working pressure angle α_w	deg	18.612	
Contact ratio ϵ	–	2.133	
Young's modulus E	GPa	206	206
Poisson's ratio ν	–	0.3	0.3
Input torque T	Nm	100	

The normal to the flank is

$$\begin{aligned} n_{VL,Rx} &= \mp \cos(\xi - \psi_b) \\ n_{VL,Ry} &= \sin(\xi - \psi_b) \end{aligned} \quad (3)$$

2.2. Modified tooth flank

The types of flank modification discussed in this study include linear tip/root relief with smoothing (profile modification) and lead crowning (lead modification). Profile grinding is used here to manufacture the modified gears for surface finishing.

2.2.1. Linear tip/root relief with a smoothing transition curve

For the purpose of profile modification, the tip and the root of the pinion are relieved linearly with an amount, C_a and C_f , respectively. In order to retain the involute property during gear meshing, a segment of the line of action having a base pitch length p_e remains non-modified, similarly to the concept of short linear relief [24]. The lengths L_{Ca} of the tip and the root relief are selected as being equal to

$$L_{Ca} = \frac{g_\gamma - p_e}{2}, \quad (4)$$

where g_γ is the length of the line of action. In order to smooth the modification curve, as shown in Fig. 2, a circular arc is added as the transition curve between the involute and linear relief region (with the length $s_{rf,a} = 0.2 L_{Ca}$). The radius of the transition arc is determined by

$$\rho_{rf,a} = \frac{0.2 \cdot L_{Ca}}{\tan(\varphi_{Cf,a}/2)} \quad (5a)$$

with

$$\varphi_{Cf,a} = \tan^{-1} \left(\frac{C_a}{0.8 \cdot L_{Ca}} \right). \quad (5b)$$

Table 2

Parameters used for flank modification and shaft misalignment.

Amount of the flank modification (mm)			
Linear tip/root relief	0.001	0.005	0.009
Lead crowning	0.01	0.015	0.02
Amount of the shaft misalignment $f_{\Sigma\beta}$ (mm)			
Shaft misalignment	0.01		0.015

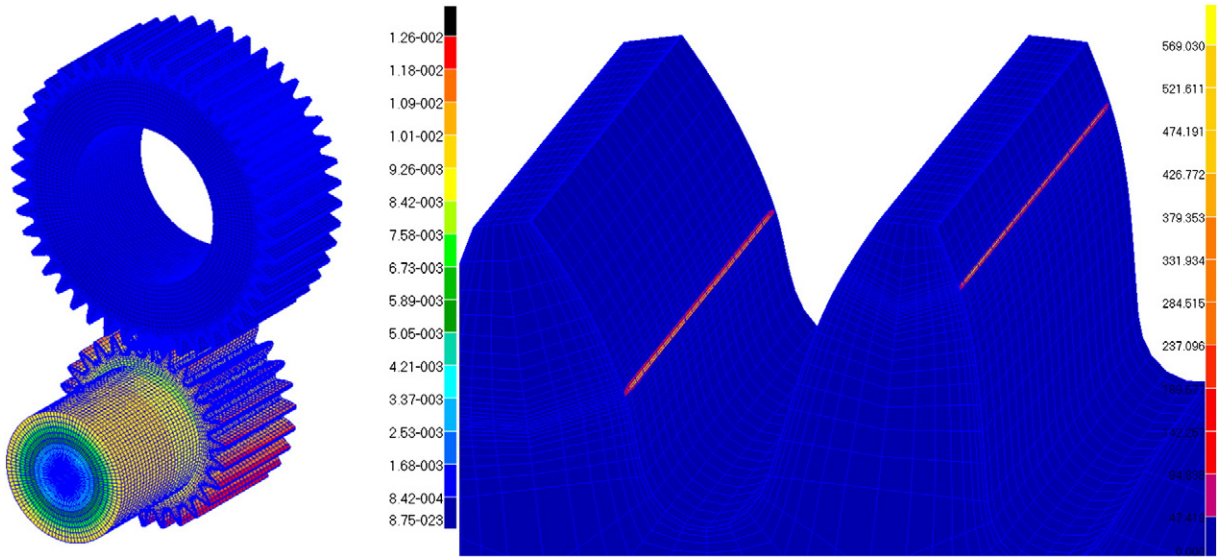


Fig. 9. FEM model and analysis result.

The relief amount C_r at a position ξ_r can be calculated according to the following regions:

$$C_r = \begin{cases} \frac{C_f}{0.8 \cdot L_{Ca}} \cdot (0.8 \cdot L_{Ca} + \xi_A - \xi_r) & \text{root linear relief, } A-R_{Lf} \\ \rho_{rf} - \sqrt{\rho_{rf}^2 - (L_{Ca} + \xi_A - \xi_r)^2} & \text{root transition relief, } R_{Lf}-R_f \\ 0 & \text{non-relief (involute), } R_f-R_a \\ \rho_{ra} - \sqrt{\rho_{ra}^2 - (\xi_r - L_{Ca} - p_e - \xi_A)^2} & \text{root linear relief, } R_a-R_{La} \\ \frac{C_a}{0.8 \cdot L_{Ca}} \cdot (\xi_r - 1.2 \cdot L_{Ca} - p_e - \xi_A) & \text{root linear relief, } R_{La}-E \end{cases} \quad (6)$$

The mathematical equations for the modified tooth profile can be derived according to the procedure mentioned above by re-writing Eq. (1) as follows:

$$\begin{aligned} x_M &= r_b \cdot \cos \xi_r + (r_b \cdot \xi_r - C_r) \cdot \sin \xi_r \\ y_M &= r_b \cdot \sin \xi_r - (r_b \cdot \xi_r - C_r) \cdot \cos \xi_r \end{aligned} \quad (7)$$

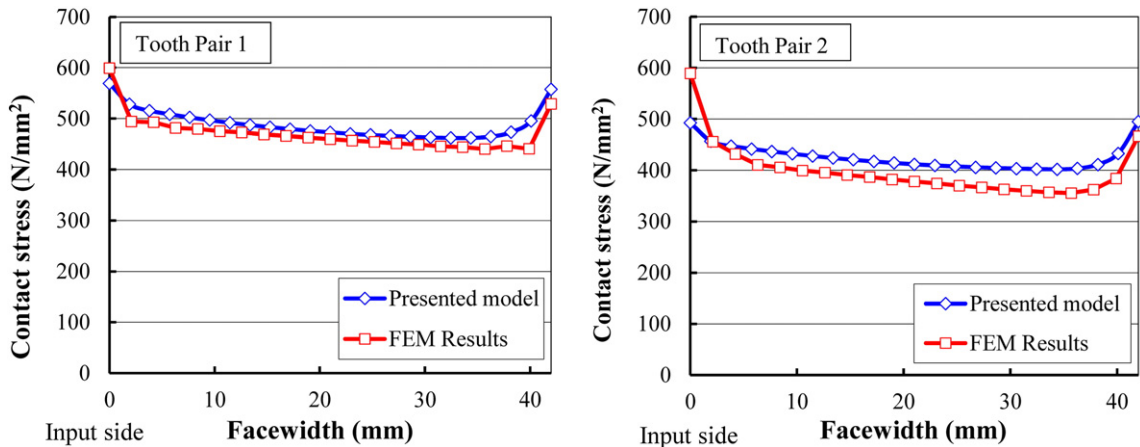


Fig. 10. Comparison of numerical analysis and FEM for both tooth pairs.

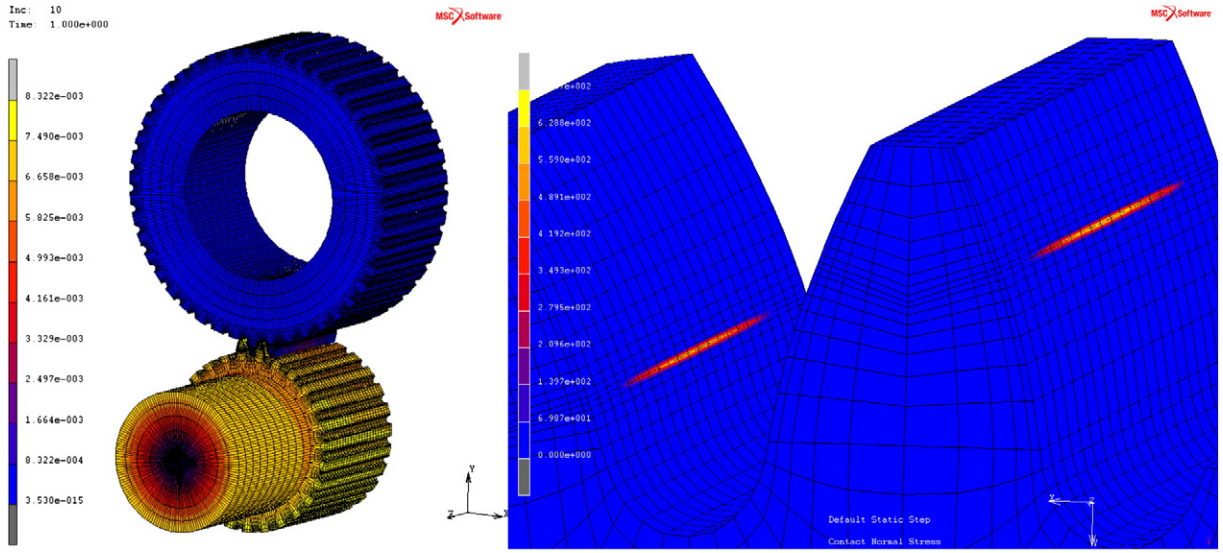


Fig. 11. FEM model with lead crowning and analysis result.

2.2.2. Lead crowning

The lead crowned flanks of the pinion are manufactured by using a profile grinder following a parabolic path, as shown in Fig. 3. For a moving distance s_C of the grinder, the crowned tooth profile of the gear is formed by the axial section profile of the grinder with an incline angle equal to γ_M . The equations for calculating the crowned flank can be derived based on the motion of the pitch point C with the distance s , i.e., the path Ω_M , shown as an equidistant curve from the parabolic center path Ω_C of the grinder. The involute profile of the grinder $\Gamma(\xi)$ represented in the coordinate system S_P (x_p, y_p, z_p) can be derived from Eq. (1) as follows:

$$\begin{bmatrix} x_{PL,R} \\ y_{PL,R} \\ z_{PL,R} \end{bmatrix} = \begin{bmatrix} \mp x_M \cdot \sin(\tau/2 - \psi_b) - y_M \cdot \cos(\tau/2 - \psi_b) \\ + x_M \cdot \cos(\tau/2 - \psi_b) \pm y_M \cdot \sin(\tau/2 - \psi_b) - r_0 \\ 0 \end{bmatrix} \quad (8)$$

$$\begin{bmatrix} n_{PL,Rx} \\ n_{PL,Ry} \\ n_{PL,Rz} \end{bmatrix} = \begin{bmatrix} \mp \cos(\xi + \tau/2 - \psi_b) \\ \sin(\xi + \tau/2 - \psi_b) \\ 0 \end{bmatrix}. \quad (9)$$

For a moving distance s along the face-width, the crowning amount t is calculated according to

$$t = \frac{4C_c}{b^2} s^2, \quad (10)$$

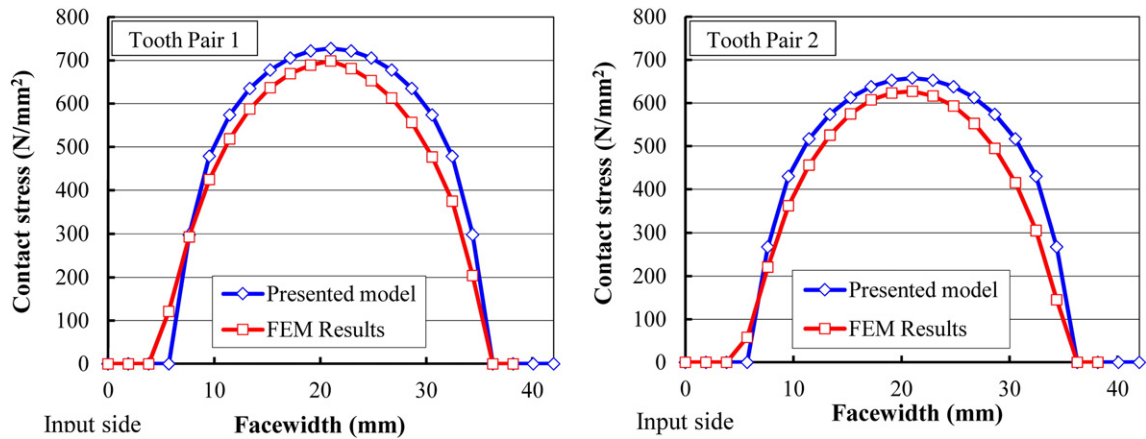


Fig. 12. Comparison of numerical analysis and FEM with lead crowning for both tooth pairs.

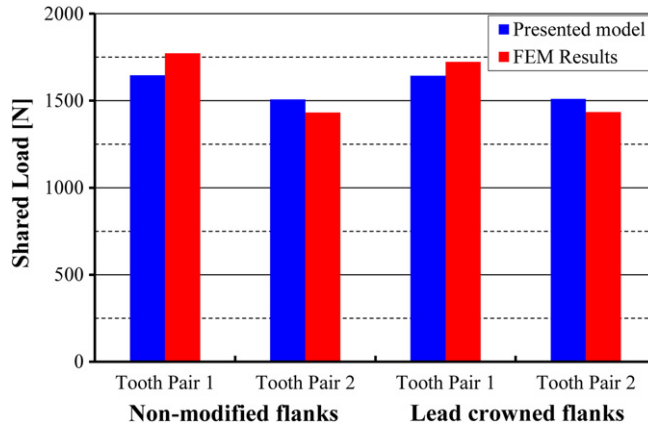


Fig. 13. Comparison of the shared loads with lead crowning for both tooth pairs.

where C_c is the crowning amount on the face-end. The angle γ_M for the normal of the parabolic path is

$$\tan \gamma_M = \frac{8C_c}{b^2} s. \quad (11)$$

The coordinates of the grinder profile in the coordinate system S_{PM} (x_{PM}, y_{PM}, z_{PM}), as in Eq. (8), are transformed to the coordinate system of the gear,

$$\begin{bmatrix} x_{CL,R} \\ y_{CL,R} \\ z_{CL,R} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \gamma_M & -\sin \gamma_M & r_0 - t \\ 0 & \sin \gamma_M & \cos \gamma_M & s \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{PL,R} \\ y_{PL,R} \\ z_{PL,R} \\ 1 \end{bmatrix} = \begin{bmatrix} x_{PL,R} \\ y_{PL,R} \cos \gamma_M - z_{PL,R} \sin \gamma_M + r_0 - t \\ y_{PL,R} \sin \gamma_M + z_{PL,R} \cos \gamma_M + s \\ 1 \end{bmatrix} \quad (12)$$

where the rotation angle γ_M around the x -axis and the translational displacements t in the y -direction and s in the z -direction are the parameters for coordinate transformation. Because the curve on the axial plane of the grinder is also a part of the crowned flank with the face-width variable s , Eq. (12) represents also the equation for the flank surface of the gear.

The normal to the crowned flank can be also derived from the coordinate transformation,

$$\begin{bmatrix} n_{cxL,R} \\ n_{cyL,R} \\ n_{czL,R} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma_M & -\sin \gamma_M \\ 0 & \sin \gamma_M & \cos \gamma_M \end{bmatrix} \begin{bmatrix} n_{pxL,R} \\ n_{pyL,R} \\ n_{pzL,R} \end{bmatrix}. \quad (13)$$

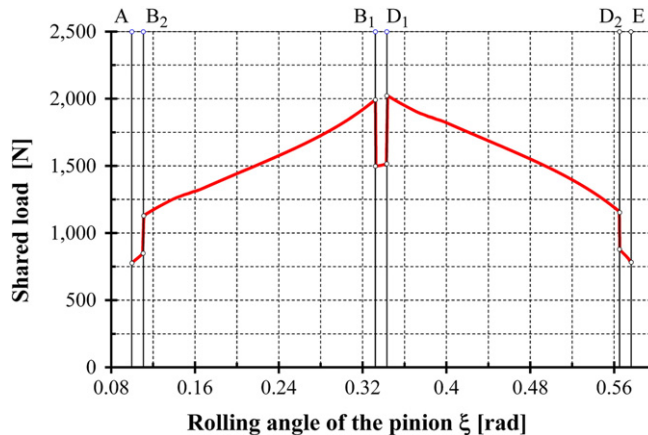


Fig. 14. Variation of loading along the normal line of action.

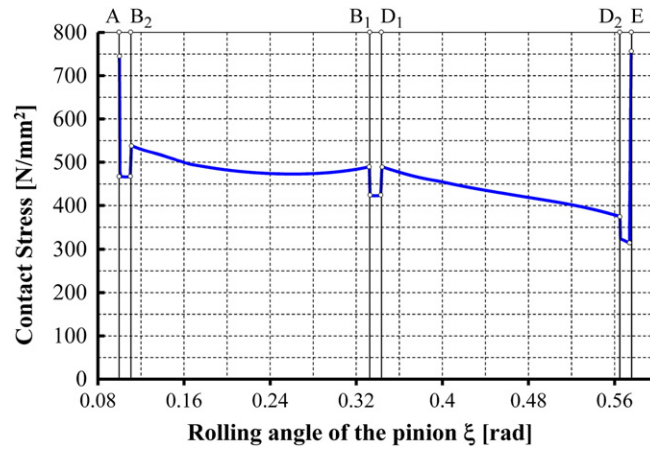


Fig. 15. Variation of contact stress along the normal line of action.

3. Analysis of gear meshing under shaft misalignment

3.1. Definition of shaft misalignment

The shafts of the paired cylindrical gears must be aligned parallel to each other. Two different alignment errors can be distinguished according to the arrangement of the shaft misalignment on the different planes [24], as shown in Fig. 4: the shaft inclination $f_{\Sigma\delta}$ on the center plane I and the shaft skew $f_{\Sigma\beta}$ on plane II which is perpendicular to plane I. In order to simplify the transformation of the coordinates, the orientation of the misaligned shaft is defined with two angles, i.e. $\Delta\theta_x$ and $\Delta\theta_y$, which are calculated from the inclination error $f_{\Sigma\delta}$ and the skew error $f_{\Sigma\beta}$ from the center distance L_{mz} of the bearings, respectively. The coordinates of each gear under ideal conditions are defined as shown in the arrangement in Fig. 4, where the coordinates $z_{1,2}$ are in the direction of the axis.

3.2. Contact of a non-modified tooth pair under shaft misalignment

An edge contact occurs when the misaligned gear 2 engages with another gear. The face-end edge of the gear 2 will be tangent to the flank of the gear 1 at the point P, as shown in Fig. 5. Two conditions must be fulfilled: (a) the position vectors $\mathbf{r}_{p1}$ and $\mathbf{r}_{p2}$ of the contact point P on the gear 1 and the gear 2 must be the same; and (b) the tangent vector $\mathbf{t}_{p2}$ at the point P to the edge must be perpendicular to the normal vector $\mathbf{n}_{p1}$ of gear 1. The tangent vector $\mathbf{t}_{p2}$ is also equal to the cross product of the normal vector $\mathbf{n}_\pi$ of the transverse section π and the normal vector $\mathbf{n}_{p2}$ of gear 2 at the contact point P. The relations can be expressed as follows,

$$\begin{aligned} \mathbf{r}_{p1} &= \mathbf{r}_{p2} \\ (\mathbf{n}_\pi \times \mathbf{n}_{p2}) \cdot \mathbf{n}_{p1} &= 0 \end{aligned} \quad (14)$$

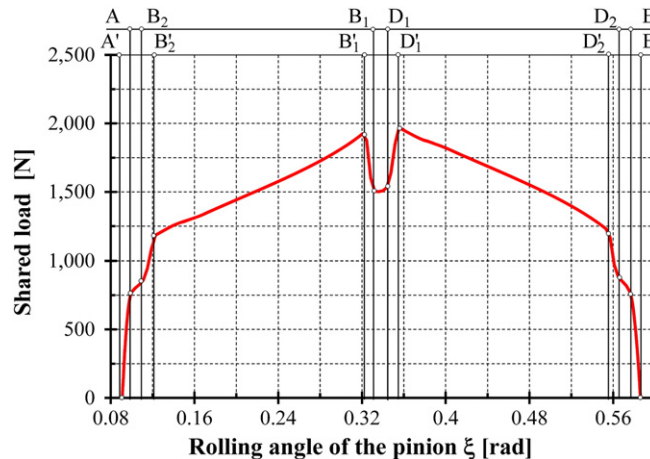


Fig. 16. Variation of the load beyond the normal line of action.

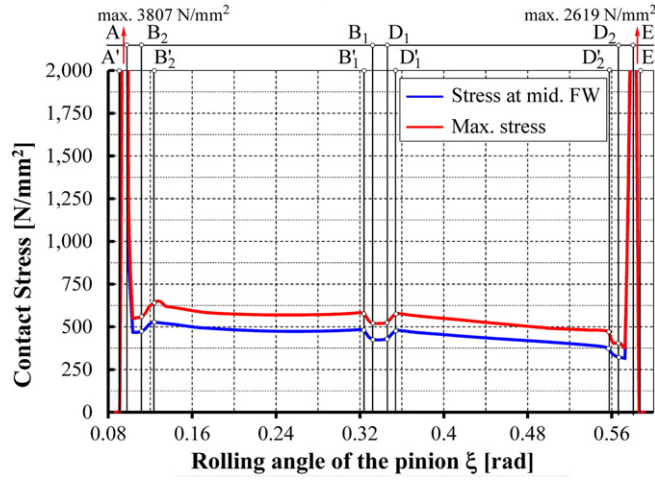


Fig. 17. Variation of contact stress beyond the normal line of action.

where the normal vector of plane π $\mathbf{n}_\pi$ is expressed as,

$$\mathbf{n}_\pi = \mathbf{M}_\Sigma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (15)$$

with

$$\mathbf{M}_\Sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\Delta\zeta_x & -\sin\Delta\zeta_x \\ 0 & \sin\Delta\zeta_x & \cos\Delta\zeta_x \end{bmatrix} \begin{bmatrix} \cos\Delta\theta_y & 0 & -\sin\Delta\theta_y \\ 0 & 1 & 0 \\ \sin\Delta\theta_y & 0 & \cos\Delta\theta_y \end{bmatrix} \quad (16)$$

and

$$\tan\Delta\zeta_x = \tan\Delta\theta_x \cos\theta_y. \quad (17)$$

The contact normal vector $\mathbf{n}_{p2}$ of gear 2 can be derived by taking the rotational angle φ_2 of gear 2 and the misaligned angle $\Delta\theta_{x,y}$, thus

$$\mathbf{n}_{p2} = \begin{bmatrix} n_{p2x} \\ n_{p2y} \\ n_{p2z} \end{bmatrix} = \mathbf{M}_\Sigma \begin{bmatrix} \cos(\xi_{p2} - \psi_{b2} + \varphi_2) \\ \sin(\xi_{p2} - \psi_{b2} + \varphi_2) \\ 0 \end{bmatrix}, \quad (18)$$

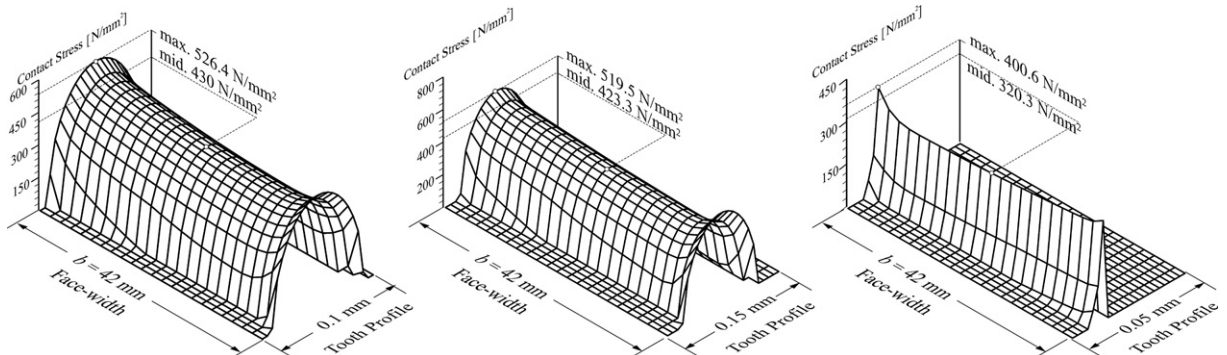


Fig. 18. Load distribution for triple tooth pair contact.

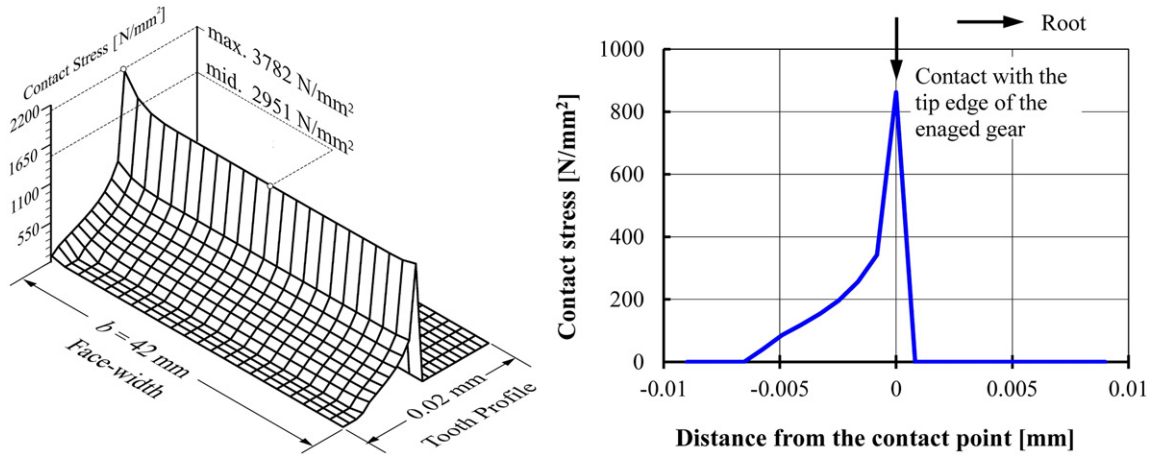


Fig. 19. Stress distribution due to tip corner contact.

and the normal vector $\mathbf{n}_{P1}$ of gear 1 is derived by taking the rotational angle ϕ_1 of gear 1 and the contact angle ξ_{P1} , while the $\psi_{b,1}$ is the half tooth thickness angle on the base circle

$$\mathbf{n}_{P1} = \begin{bmatrix} \cos(\xi_{P1} - \psi_{b,1} + \phi_1) \\ \sin(\xi_{P1} - \psi_{b,1} + \phi_1) \\ 0 \end{bmatrix}. \quad (19)$$

3.3. Contact of tooth pairs with flank modification

In order to avoid the edge contact of gear pairs in the presence of misalignment of the gear axis, the flanks of one gear are usually crowned in the lead direction. Thus gear meshing is transformed from line contact to point contact. It is necessary to determine the shifting of the contact points and the transmission errors of the gear pair under the misaligned condition.

In general, the contact point of two engaged flanks and the corresponding rotation angles of the gears at every instant are determined with a set of five independent equations, based on the conditions of continuous tangency of the tooth surfaces [25]. In order to simplify the calculation, the relation of coincidence of the normal vectors can also be solved independently by using the characteristics of the involute geometry [8]. According to the relation, as shown in Fig. 6, the contact normal $\mathbf{n}_p$ is tangential to the base cylinder of gear 2 and perpendicular to the involute generator of the non-modified flank Γ_2 . This geometrical condition is equivalent to that the normal $\mathbf{n}_p$ at point P (considered as a line) is skewed to the axis with an angle and an offset. It can be represented from the line geometry by the scalar product of the dual vectors. Namely, with the dual normal $\hat{\mathbf{n}}_p$, the dual vector of the axis $\hat{\mathbf{A}}_2$ and the dual angle $\hat{\vartheta}$ the equation is valid,

$$\hat{\mathbf{n}}_p \cdot \hat{\mathbf{A}}_2 = \cos \hat{\vartheta}. \quad (20)$$

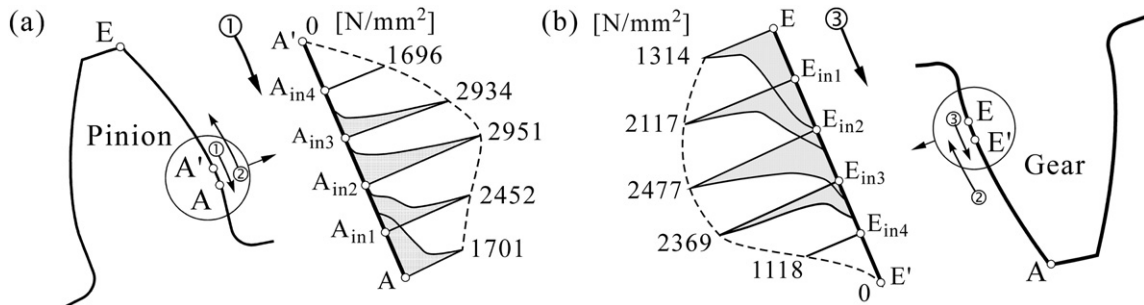


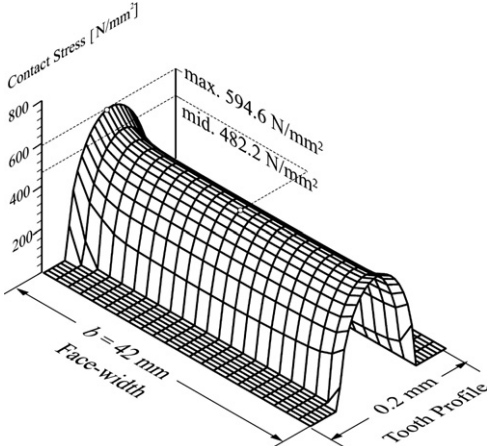
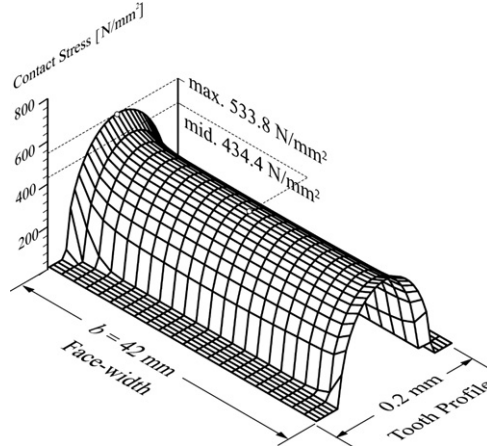
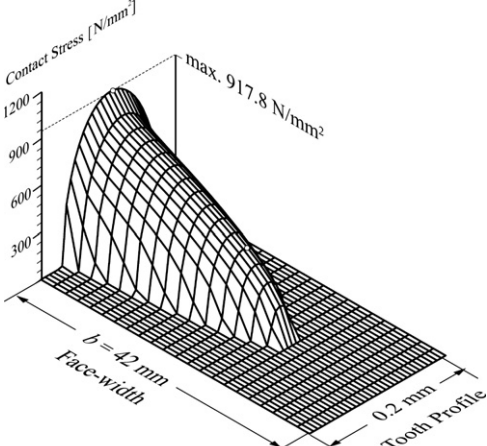
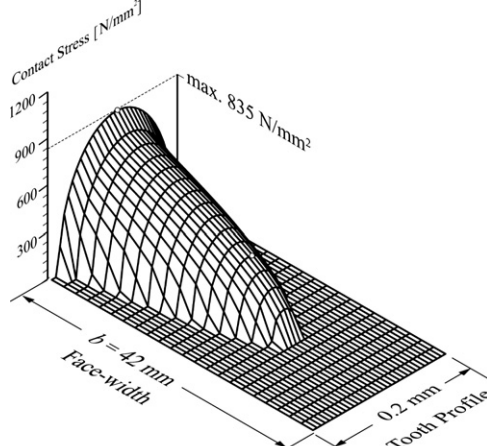
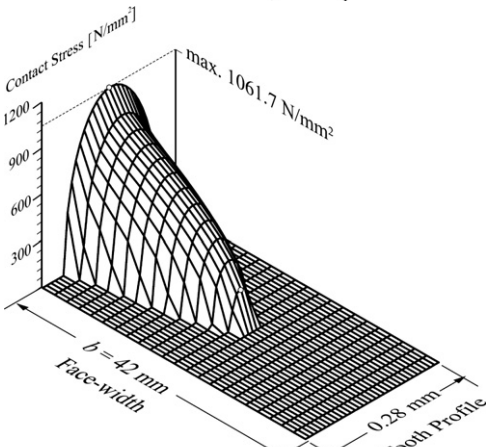
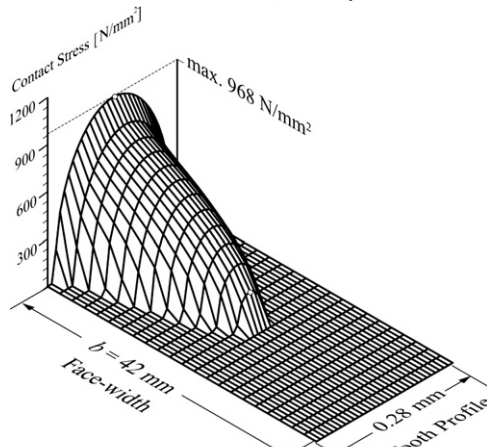
Fig. 20. Variation of the stress distribution due to tip corner contact on the flanks.

Because the involute generator is parallel to the axis, the dual angle $\hat{\vartheta}$ is equal to $\pi/2 + \varepsilon r_b$, Eq. (20) can be further rewritten as

$$\hat{\mathbf{n}}_p \cdot \hat{\mathbf{A}}_2 = \cos(\pi/2) - \varepsilon r_{b2} \cdot \cos(\pi/2) = 0 - \varepsilon r_b. \tag{21}$$

Eq. (21) represents two independent equations for two unknown variables of point $P(\xi_1, s)$ on the modified flank.

Table 3
Contact stress distribution under different misalignment errors.

f_{2p} [mm]	Tooth pair 1	Tooth pair 2
0		
0.01		
0.015		

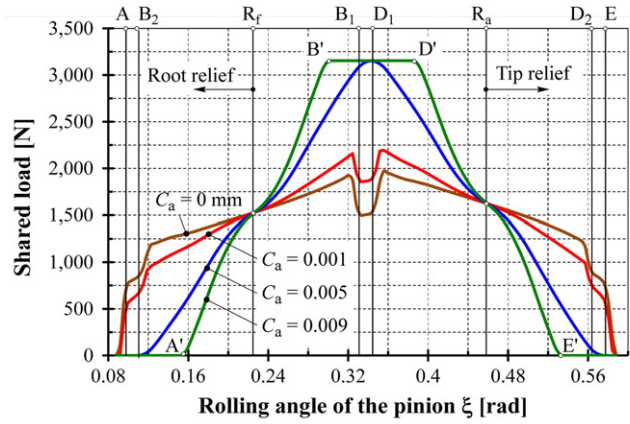


Fig. 21. Variation of loading with different amounts of linear short tip and root relief.

According to the dual vector definition, the dual normal vector $\hat{\mathbf{n}}_p$ is expressed as

$$\hat{\mathbf{n}}_p = \mathbf{n}_p + \varepsilon \mathbf{r}_p \times \mathbf{n}_p = \begin{bmatrix} n_{px} \\ n_{py} \\ n_{pz} \end{bmatrix} + \varepsilon \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} \times \begin{bmatrix} n_{px} \\ n_{py} \\ n_{pz} \end{bmatrix} = \begin{bmatrix} n_{px} \\ n_{py} \\ n_{pz} \end{bmatrix} + \varepsilon \begin{bmatrix} y_p \cdot n_{pz} - z_p \cdot n_{py} \\ -x_p \cdot n_{pz} + z_p \cdot n_{px} \\ x_p \cdot n_{py} - y_p \cdot n_{px} \end{bmatrix}. \quad (22)$$

The normal vector $\mathbf{n}_p$ is determined from Eq. (13) with a given rotation angle φ_1 of the pinion and the dual vector $\hat{\mathbf{A}}_2$ is

$$\mathbf{A} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (23)$$

With Eqs. (22) and (23) the two independent equations from Eq. (21) are

$$\begin{aligned} n_{pz} &= 0, \\ x_p \cdot n_{py} - y_p \cdot n_{px} &= -r_b. \end{aligned} \quad (24)$$

The coordinates of the contact point P, also the surface variables, and the rotation angle of gear 2 can later be determined using the two solved surface variables, ξ_1 and s , of the corresponding pinion.

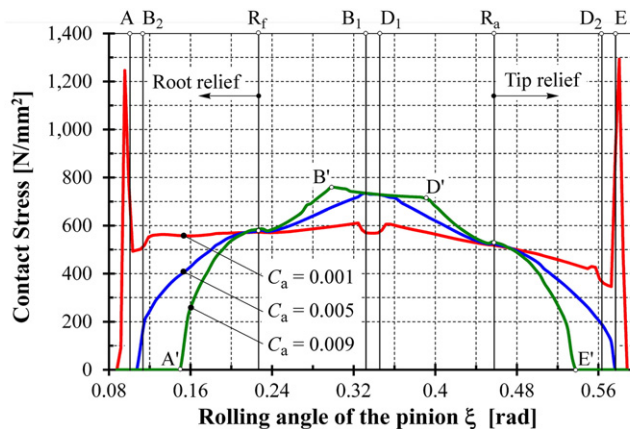
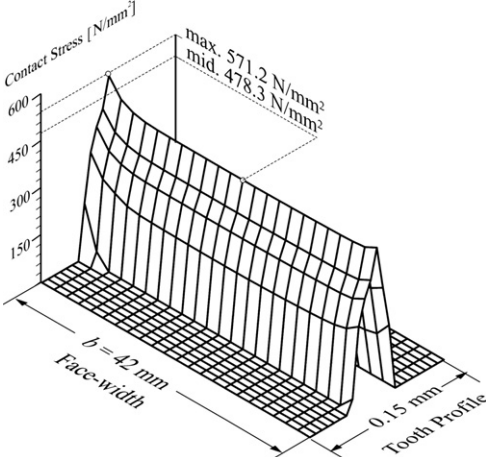
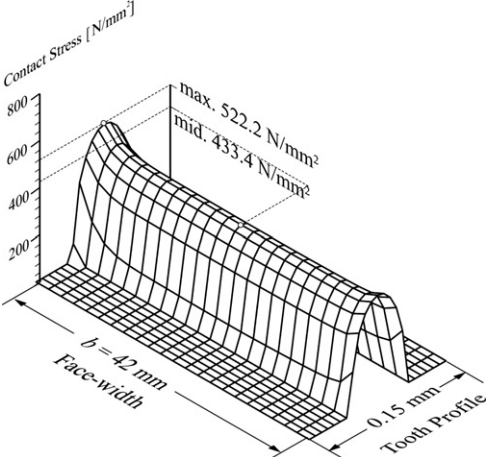
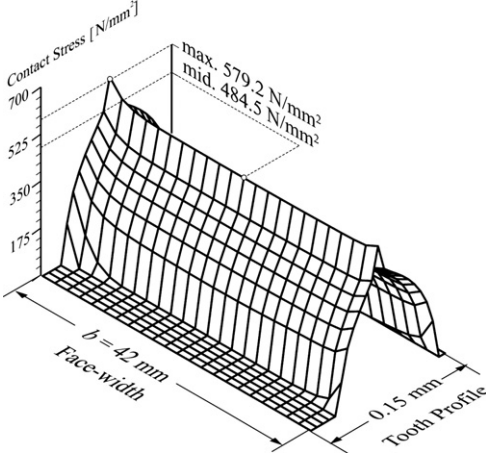
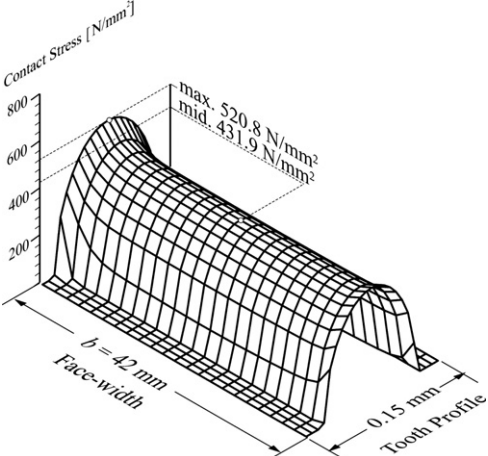
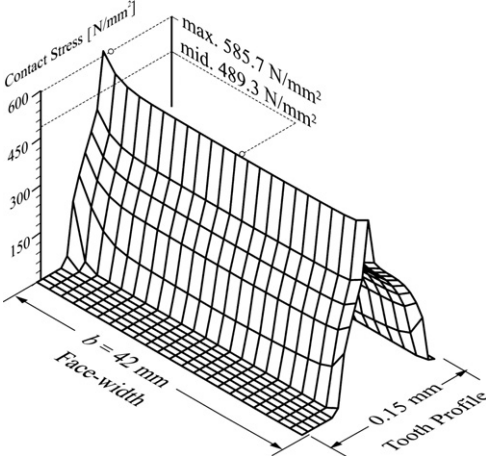
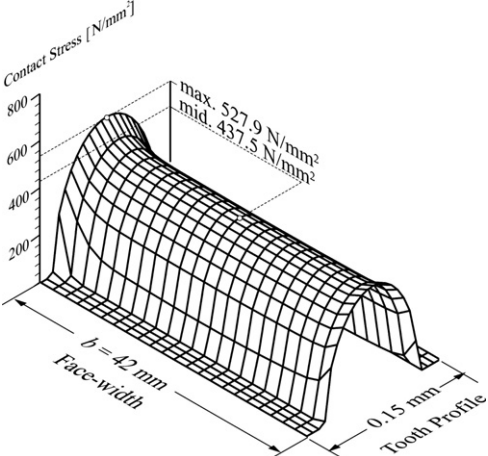


Fig. 22. Variation of stress with different amounts of linear short tip and root relief.

Table 4
Stress distribution at the beginning of the tip relief with different amounts of relief.

The amount of relief C_a (mm)	Tooth pair 1	Tooth pair 2
0.001		
0.005		
0.009		

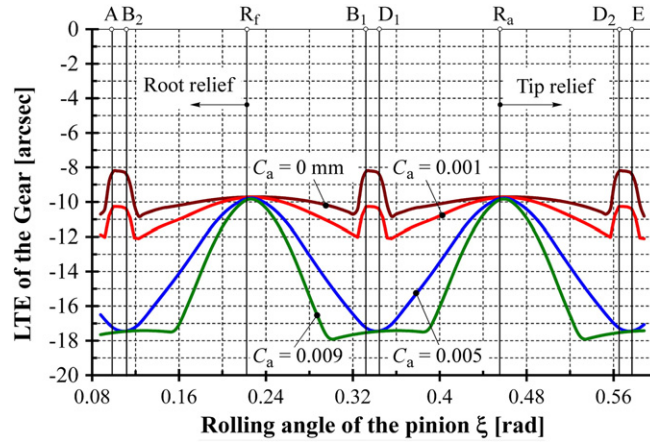


Fig. 23. Loaded transmission errors after different amounts of relief.

4. Loaded tooth contact analysis

4.1. Basic LTCA model

4.1.1. Fundamentals of the LTCA model

The model of loaded tooth contact analysis (LTCA) for multiple tooth-pairs is based on the influence coefficient method considering the relation of the deformation and the loading on a contact region. For a tooth pair, as shown in Fig. 7(a), the following relations hold between the approaching distance δ of the tooth pair, the deformation w_{Q_1, Q_2} and the separation distance h at points Q_1 and Q_2 :

$$h + w_{Q_1} + w_{Q_2} = \delta, \text{ if points } Q_1 \text{ and } Q_2 \text{ are in contact;} \quad (25)$$

$$h + w_{Q_1} + w_{Q_2} > \delta, \text{ if points } Q_1 \text{ and } Q_2 \text{ are beyond contact.} \quad (26)$$

In order to determine the deformation of each point on the engaged flanks, the contact region, as shown in Fig. 7(b), is discretized into a lot of small units with the same area s ($s = l_A \times l_B$). The total elastic deformation w_k at point P_k can be expressed as a summation of the deformation caused by all the contact stresses p_j ($j = 1, 2, \dots, n$), which are determined with influence coefficients f_{kj} , i.e.

$$w_k = \sum_{j=1}^n f_{kj} p_j. \quad (27)$$

If the contact region is discretized into n units, n equations for the deformation–displacement relations Eq. (27) and an equation of load equilibrium,

$$F = \sum_{j=1}^n s \cdot p_j, \quad (28)$$

can be obtained for solving the n unknown contact stresses p_j and the unknown approaching distance δ . This can be further expressed in the form of a matrix equation [8,13]:

$$\begin{bmatrix} \mathbf{A} & -\mathbf{I}_{n \times 1} \\ s\mathbf{I}_{1 \times n} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{P} \\ \delta \end{bmatrix} = \begin{bmatrix} -\mathbf{H} \\ F \end{bmatrix}, \quad (29)$$

Table 5

Transmission errors with different amounts of linear relief.

No.	Relief amount [mm]	Misalignment [mm]	Max. LTE [arcsec]	Min. LTE [arcsec]	Peak–peak value [arcsec]
1	0	0	0	0	0
2	0.001	0	−9.7216	−12.099	2.3774
3	0.005	0	−9.7772	−17.45	7.6728
4	0.009	0	−9.8304	−17.9181	8.08767

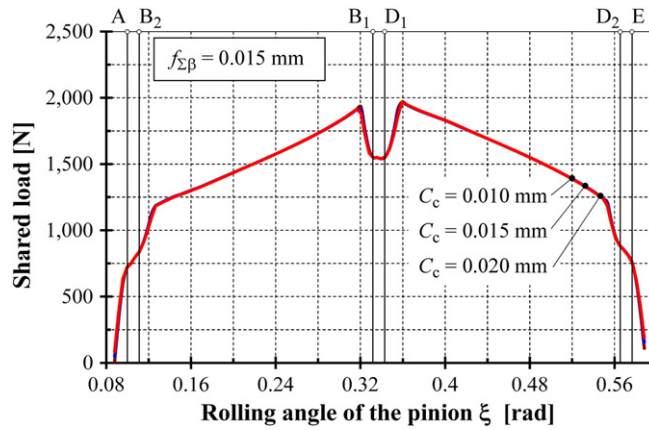


Fig. 24. Variation of loads due to different crowning amounts under the same shaft misalignment.

or

$$\mathbf{AP} = \mathbf{H}. \quad (30)$$

Eqs. (29) or (30) can be effectively solved with the aid of the LU-decomposition method. Because the initial region for the calculation is not the actual contact area of the tooth pair, some of the calculated pressure p_j will be negative. This means that the units with a negative pressure are out of contact under loading. Thus matrices $\mathbf{A}$ and $\mathbf{H}$ are reduced by excluding the units with a negative pressure. The solution process is iterated until the convergence condition is fulfilled, that no negative pressure is calculated [8,13]. The contact pattern of the flanks can be also simulated by the calculated area of convergence, as shown in Fig. 7(b).

4.1.2. Separation distance between engaged flanks

The separation distance h between the discrete points Q_1 and Q_2 on the flanks can be calculated using Eq. (12) with the aid of the geometric arrangement of the flank surfaces of the gear pair. A coordinate system $S_y(x_y, y_y, z_y)$ is introduced. The origin O_y is the contact point of an engaged tooth pair, as in Fig. 7(b). The z -axis of the coordinates is the contact normal and directed towards the flank of gear 2. The x -axis is oriented in the direction of the major axis of contact ellipse. The tooth surface points are transformed into the coordinate system S_y . For each given discrete point $P_k(x_k, y_k)$ on the common tangent plane, the corresponding z -coordinate z_{1k} or z_{2k} for one of the two engaged flanks can be determined with the aid of a non-linear numerical method. The separation distance h_k for discrete point P_k can be obtained according to the relation:

$$h_k = z_{1k} - z_{2k}. \quad (31)$$

According the proposed LTCA method, the separation distances h affect the contact stress strongly. Not only the flank surface with or without flank modification, but also the arrangement of the gear axes under aligned or misaligned conditions, can be also involved in the calculation of separation distances h in the proposed model.

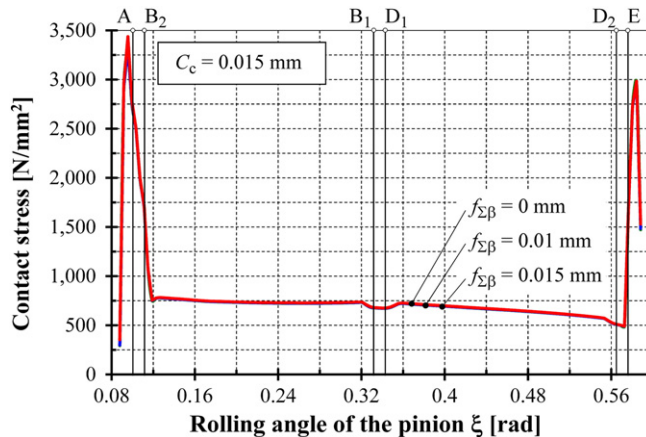


Fig. 25. Variation of middle contact stress with different amounts of shaft misalignment.

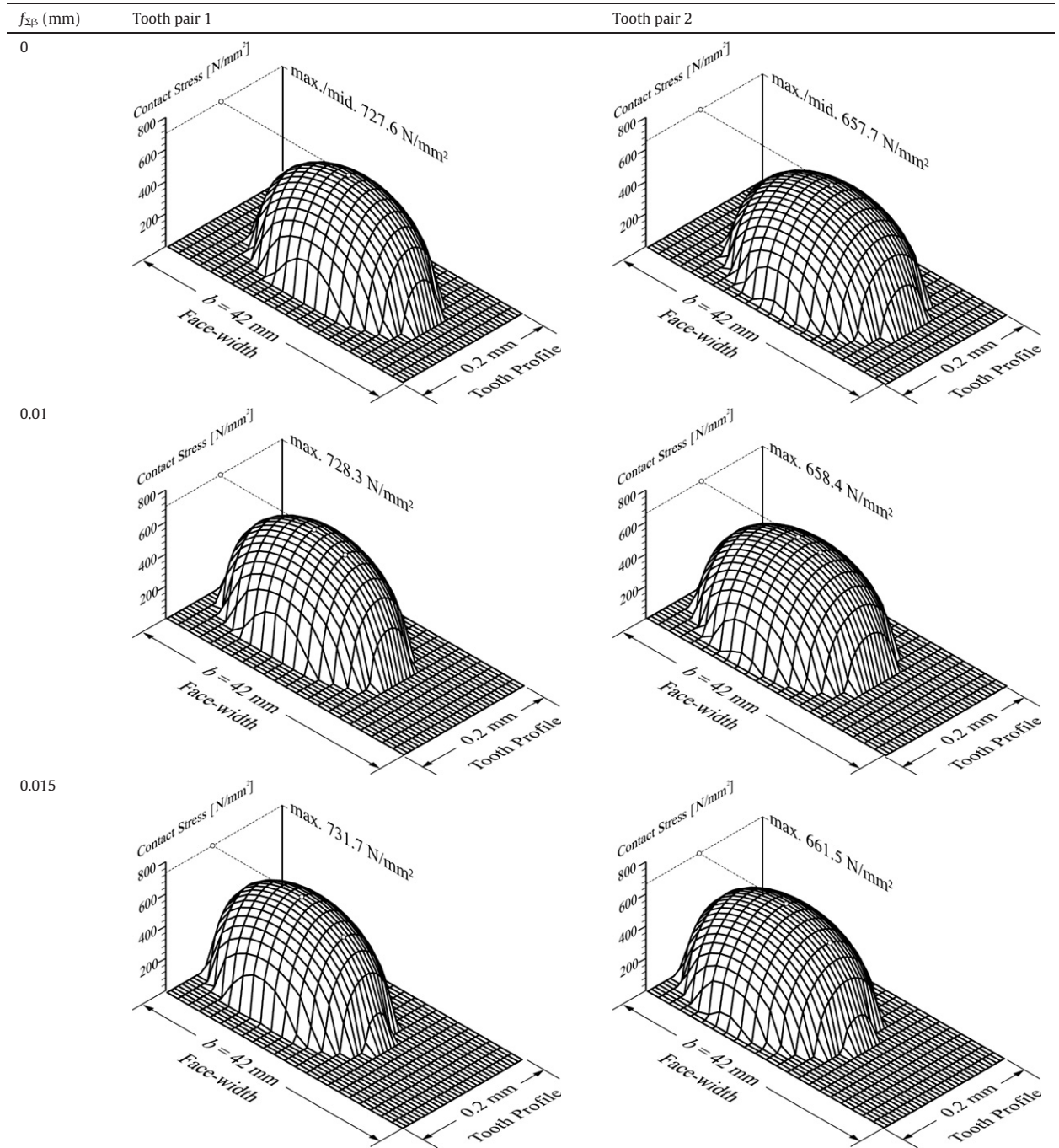
4.1.3. Influence coefficients

The influence coefficients in the basic LTCA model proposed in [8,13] are used for the deformations due to the Hertzian contact, tooth bending and shear loading. With the assumption of superposition of deformations, each influence coefficient f_{kj} is equal to the summation of the mentioned loaded deformation,

$$f_{kj} = f_{H,kj} + f_{B,kj} + f_{S,kj}. \quad (32)$$

Table 6

Influence of the amount of shaft misalignment on the stress distribution on flanks.



The influence coefficient f_H for Hertzian contact deformation is determined based on the Boussinesq half-space force–displacement relations [26]. The coefficient f_H for tooth bending is determined under consideration the deflection variation along the face-width of the concentrated loaded tooth [27–29], while the coefficient for shear deformation is determined according the general mechanics of material [30]. More details can be found in [8,13].

4.1.4. Model for multiple tooth-pair contact

In the case of triple tooth-pair contact, the matrix Eq. (29) can be further expanded to

$$\begin{bmatrix} \mathbf{A}_1 & 0 & 0 & -\mathbf{I}_{n_1 \times 1} \\ 0 & \mathbf{A}_2 & 0 & -\mathbf{I}_{n_2 \times 1} \\ 0 & 0 & \mathbf{A}_3 & -\mathbf{I}_{n_3 \times 1} \\ s_1 \mathbf{I}_{1 \times n_1} & s_2 \mathbf{I}_{1 \times n_2} & s_3 \mathbf{I}_{1 \times n_3} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \mathbf{P}_3 \\ \delta \end{bmatrix} = \begin{bmatrix} -\mathbf{H}_1 \\ -\mathbf{H}_2 \\ -\mathbf{H}_3 \\ F \end{bmatrix}, \quad (33)$$

under the condition that the loaded deformation of each individual tooth pair is independent of each other.

4.2. Extended LTCA model considering the influence of twist deformation

4.2.1. Influence of the shaft twisting on the load–deformation relation of the tooth pairs

Eq. (33) of the proposed basic LTCA model is only valid for the loaded condition considering tooth bending, Hertzian contact and shear deformation of the engaged teeth, whereby the loading on one tooth pair will not cause the displacement of any discrete point on another tooth pair. In reality, the influence of the twist of the shafts due to the transmitted torque on the contact stress cannot be ignored. A geared shaft under torque loading is considered (Fig. 8). The reaction force $F_{II,i}$ of the applied torque acting at point $P_{II,i}$ on tooth pair II will cause twist deformations at point $P_{I,k}$ on tooth pair I and $P_{II,k}$ of tooth pair II at the same time. This means that the load–deformation relations among the engaged tooth-pairs are affected by each other. Eq. (33) is no longer valid for such deformation at point $P_{I,k}$ on flank I. Therefore, the influence of the load $F_{II,i}$ acting on flank II should be included in the equation.

In the case of triple tooth-pair contact, the relation for deformation at point k on tooth pair I, Eq. (27), should be rewritten with the relation $F_{I(II, III),i} = s_{1(2, 3)} \cdot p_{I(II, III),i}$ as follows:

$$w_{I,k} = \sum_{i=1}^{n_1} f_{I-I,ki} \cdot p_{I,i} + \sum_{i=1}^{n_2} f_{I-II,ki} \cdot p_{II,i} + \sum_{i=1}^{n_3} f_{I-III,ki} \cdot p_{III,i}. \quad (34)$$

The matrix Eq. (33) for triple tooth-pair contact can be further expressed as

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} & -\mathbf{I}_{n_1 \times 1} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \mathbf{A}_{23} & -\mathbf{I}_{n_2 \times 1} \\ \mathbf{A}_{31} & \mathbf{A}_{32} & \mathbf{A}_{33} & -\mathbf{I}_{n_3 \times 1} \\ s_1 \mathbf{I}_{1 \times n_1} & s_2 \mathbf{I}_{1 \times n_2} & s_3 \mathbf{I}_{1 \times n_3} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \mathbf{P}_3 \\ \delta \end{bmatrix} = \begin{bmatrix} -\mathbf{H}_1 \\ -\mathbf{H}_2 \\ -\mathbf{H}_3 \\ F \end{bmatrix}. \quad (35)$$

The matrices of the influence coefficients $\mathbf{A}_{ij}$ ($i \neq j$) in Eq. (35) are only used for the twist deformation, while the matrices $\mathbf{A}_{ii}$ consist of the coefficients in the basic model, i.e., considering contact, bending and shear deformation of the tooth pairs, in addition to the coefficients for twist deformation.

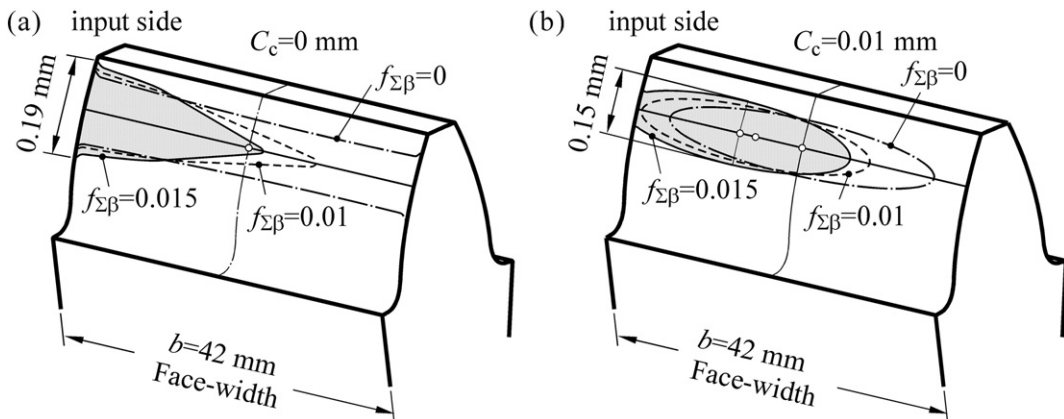


Fig. 26. Comparison of contact patterns of the gear pair with and without lead crowning in the presence of various shaft misalignments.

Table 7

Loaded transmission errors with the influence of crowning and misalignment.

No.	Crowning [mm]	Misalignment [mm]	Max. LTE [arcsec]	Min. LTE [arcsec]	Peak-to-peak value [arcsec]
1	0	0	−8.2	−10.7	2.5
2	0.01	0	−19.1	−22.5	3.4
3		0.01	−38.0	−41.4	3.4
4		0.015	−46.9	−50.4	3.5
5	0.015	0.01	−40.8	−44.6	3.8
6		0.015	−49.9	−53.7	3.8
7	0.02	0.01	−42.9	−46.9	4.1
8		0.015	−52.1	−56.3	4.2

4.2.2. Influence coefficient for the twist deformation

The influence coefficient for the twist deformation is determined based on the relation of the applied torque and the twist deformation of the shaft

$$\kappa = \frac{T \cdot L}{J_t \cdot G}. \quad (36)$$

A hollow geared shaft is loaded under a torque T and a reaction force $F_{ll,i}$ at point $P_{ll,i}$, as shown in Fig. 8. The twist angles $\kappa_{l,IIIk}$ of point P_{lk} and P_{llk} on flank I and II, respectively, will be the same, provided both points are located at the same axial distance relative to point $P_{ll,i}$. The deformation can be thus determined under two conditions: the shaft segment is under torsional stress or is without torsional stress. The deformation w_k of the flank point P_k caused by the reaction force $F_{ll,i}$ along the line of action of the gear pair is equal to $r_b \cdot \kappa_{l,IIIk}$.

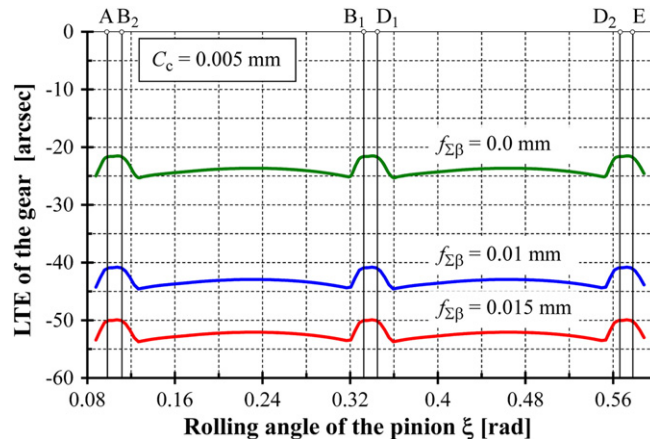
Flank point P_m of any tooth pair having a distance x_m and a reaction force F_i acting at point P_i having a distance x_i are considering (Fig. 8). According to the definition, $w_m = f_{ki} \cdot s \cdot p_i$, and the influence coefficient can be determined according to the two afore-mentioned conditions.

(1) When the shaft segment is under torsional stress, i.e., $x_m \leq x_i$, the influence coefficient f_{mi} is equal to

$$f_{m,i} = x_m \cdot \frac{2 \cdot s \cdot r_b \cdot r_f}{\pi \cdot (r_f^4 - r_i^4) \cdot G}. \quad (37)$$

(2) When the shaft segment is without torsional stress, i.e., $x_m > x_i$, the influence coefficient f_{mi} is equal to

$$f_{m,i} = x_i \cdot \frac{2 \cdot s \cdot r_b \cdot r_f}{\pi \cdot (r_f^4 - r_i^4) \cdot G}. \quad (38)$$

**Fig. 27.** Loaded transmission errors given with different amounts of misalignment.

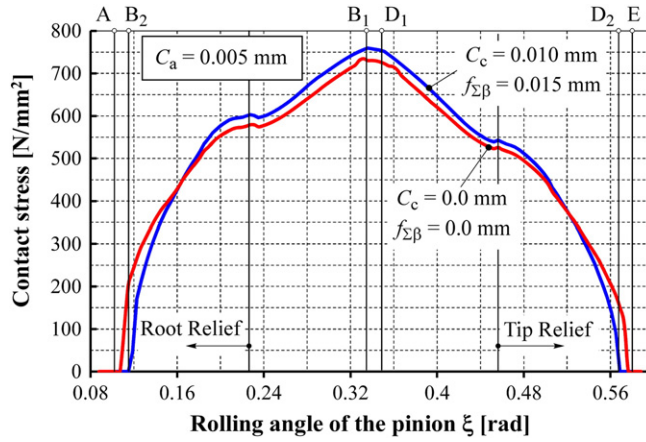


Fig. 28. Variation in contact stress of crowned gears with shaft misalignment compared with non-crowned gears.

4.3. Unloaded and loaded transmission errors

The transmission error can be analyzed under the loaded condition or without loading. In general, no unloaded transmission error will be present in a gear pair under ideal conditions. However, in the case of a gear pair having assembly/manufacturing errors and/or flank modifications, the unloaded transmission error (UTE) will exist, and is define as,

$$UTE = \Delta\varphi_2 = (\varphi_2 - \varphi_{20}) - \frac{z_1}{z_2} \cdot (\varphi_1 - \varphi_{10}), \quad (39)$$

where $z_{1(2)}$ and $\varphi_{1(2)}$ are the tooth number and the rotation angle of driving gear 1 and driven gear 2, respectively.

On the other hand, the transmitted angular relation of the gear pair under loading doesn't remain constant due to the variation of the meshing stiffness. The loaded transmission error (LTE) can be defined as the resultant deviation of the transmitted angle $\Delta\varphi_2$ due to the compliance angle φ_δ , i.e.,

$$LTE = \Delta\varphi_2 = (\varphi_2 - \varphi_{20}) - \frac{z_1}{z_2} \cdot (\varphi_1 - \varphi_{10} + \varphi_\delta) = UTE - \frac{z_1}{z_2} \cdot \varphi_\delta. \quad (40)$$

The compliance angle φ_δ can be obtained from the calculated approaching displacement δ from the proposed LTCA model, Eq. (24), by using the relation

$$\varphi_\delta = \frac{\delta}{r_{b1}}. \quad (41)$$

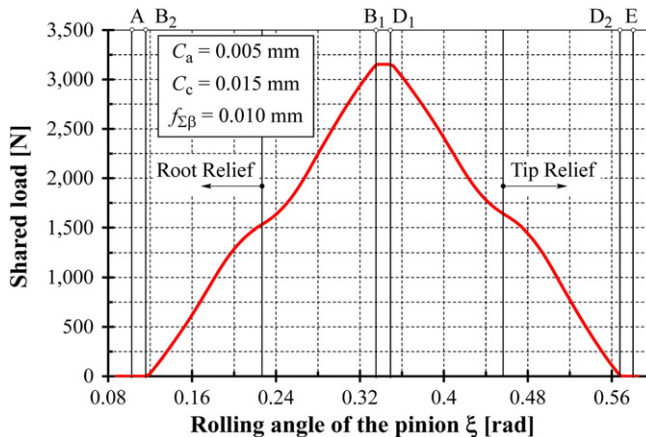


Fig. 29. Variation in shared loads of crowned gears with shaft misalignment.

5. Study cases

The contact characteristics of a spur gear pair with a high contact ratio 2.133 are studied. The corresponding gearing data for the spur gears are listed in Table 1. The gears have a high tooth profile and are paired as a negative profile-shifted drive. The contact stress and the load sharing of the gear pair are analyzed with consideration of the misalignment of the shafts and the flank modification. The flank modification consists of lead crowning and linear tip/root relief. The amounts of flank modification and shaft misalignment for analysis (Table 2) are selected according to DIN 3960 [23] and Niemann [24], respectively.

5.1. Compared with FEM analysis

The tooth contact stresses of the gear pairs with and without lead crowning are analyzed using the proposed LTCA model and the FEM software MSC.Marc. The gearing data are described in Table 1. The tooth pairs under an input torque of 100 Nm applied on the input shaft of the pinion engage each other as double tooth-pair contacts. The rotation angle of the pinion is 0.264 rad. The type of element used for the FEM is Hex8 (6 faces and 8 nodes). No special elements are required to be placed at the points of contact, because the special algorithm of the FEM software, i.e., Solver Constrain Method, can automatically detect nodes in contact and solve the contact problem. The Young's modulus and Poisson's ratio are 206 GPa and 0.3, respectively. The pinion is located axially and radially, and can only rotate freely about its axis, i.e., the three directions of translation and two radial directions of rotation are defined as fixed. The driven gear is fixed in all the six directions.

At first the non-modified gear pair is considered for comparison. The FEM model and the analysis results are shown in Fig. 9. A comparison of results from the FEM software and the LTCA model, as illustrated in Fig. 10, shows good agreement. The contact stress obtained by FEM is slightly smaller than that calculated by the numerical analysis. The stress of tooth pair 1 is larger than that of tooth pair 2 due to the meshing stiffness. The contact stress along the face-width of the two tooth pairs near the face-end where the torque is applied is enlarged because the twist deformation near this face-end is larger than another.

On the other hand, the analysis results of the lead-crowned gear pair with an amount of 0.01 mm obtained from FEM are shown in Fig. 11. The variation of the contact stress along the major axis of the contact ellipse calculated by the proposed model has similar trends of that by FEM, as shown in Fig. 12. The value of the contact stress calculated by FEM is also slightly smaller than the numerical analysis, similar to the case of non-modified gear pair. The influence of the twist deformation on the ellipse-type stress distribution is not clearly to identify. It is different from the case of the non-modified gears.

In order to evaluate the analysis results, the shared loads of the two tooth pairs calculated by the two models are compared, as the bar chart in Fig. 13 shows. The results calculated by the two methods reveal that the load acting on tooth pair 1 is larger than tooth pair 2 due to the different mesh stiffness. The shared load on tooth pair 1 calculated by FEM is larger than that by the proposed approach, although the calculated contact stress is smaller. These differences of the results obtained by the two methods come from two considerations:

- Some influences of the deformation, such like the foundation deformation of the gear blank, are not considered in the proposed LTCA approach.
- The mesh quality of FEM, such like the precise location and the quantity of the nodes, affects the results strongly.

However, the comparative results reveal still the feasibility of the proposed LTCA approach for analyzing the contact characteristics of the spur gears.

5.2. Case 1: tooth contact under ideal assembly conditions

The contact characteristics of a spur gear pair without flank modification under ideal assembly conditions are analyzed. The influence of the tip corner contact of engaged tooth pairs on the contact characteristics is also considered in this case. The analysis results are also compared with the FEM results to verify the proposed LTCA approach.

5.2.1. Stress and load sharing without considering tip corner contact

The shared load of an individual tooth pair during gear meshing is influenced by the number and also the stiffness of the contacting tooth-pairs. First, it should be noted that the tip corner contact is not considered in the analysis. The calculated results of the variation in the shared load can be identified in Fig. 14. Within the normal line of action AE, the regions B_2B_1 and D_1D_2 represent the double tooth-pair contact, respectively, and the other regions triple tooth-pair contact. The shared load of the individual tooth pair varies rapidly at points A, B_2 , B_1 , D_1 , D_2 and E, where the number of tooth pairs in contact changes. Because of the varied mesh stiffness, there is also variation in the shared load in regions having the same number of contacting tooth pairs. The relative maximum load occurs at the middle of the gear meshing, here at position D_1 .

The variation of the corresponding maximum contact stress is illustrated in Fig. 15. There is a rapid increase in stress at the positions near where the contact begins and ends (points A and E) due to concentration of stress at the tooth tip edge. The contact stress of the tooth pair is a little larger in region B_2B_1 (shortly after contact begins) than that in region D_1D_2 (shortly before contact ends) because of the variation in the curvature relation between the pinion and the gear.

5.2.2. Stress and load sharing considering the tip corner contact

In actual gear meshing conditions, tip corner contact due to compliance of the gear teeth cannot be avoided. The influence of tip corner contact is considered in shared loads, as shown in Fig. 16. Variation is smooth without any jump, because the loads in regions $B_1'B_1$ and $D_2'D_2$, and $B_2'B_2$ and $D_1'D_1$ are further shared by the tooth contact in the region $A'A$ and $E'E$, located beyond the normal line of action. The actual line of action and the region of triple tooth-pair contact are enlarged correspondingly.

Although it seems that the loading conditions can be improved by additional contact with the tooth tip edge, this is not a benefit in practical applications due to the occurrence of non-Hertzian contact. Fig. 17 shows the analysis results of the variation of the contact stresses. The maximum stress does not occur at positions A or E (see Fig. 15), but in the regions of tip corner contact. The contact stress within the normal line of action varies similarly to the ideal case, but with no jump.

The distributions of the contact stress on the flanks under triple tooth pair contacts within the normal line of action are illustrated in Fig. 18. The value of the stress decreases from the face-end near the position where the torque is applied to the other end, due to the twist deformation of the shaft. The stress concentration on both of the face-ends can also be found in the simulation results. Although the tooth pair 2 has the largest shared load, the contact stress is a little smaller than tooth pair 1 due to the curvature relation. It can be also identified from the difference of the contact patterns in Fig. 18, that the area of tooth pair 2 is larger than that of tooth pair 1. The tendency of the curvature relation is also valid for tooth pair 3. The contact stress still remains at a slightly higher value with the smallest contact pattern, although the shared load is the smallest.

The contact stress in the regions AA' and EE' beyond the normal line of action is enlarged due to the interference of tooth engagement, although the shared load is smaller than that for the other two tooth pairs. Fig. 19 shows the distribution of the contact stress on the flank of the pinion at the contact position shortly before point A. A typical non-Hertzian contact can be identified from the concentrated stress arising from the tip corner contact. The location of the contact points on the flank of the pinion changes from the actual contact begin point A' to the normal beginning contact point A in the direction towards the tooth root; see to Fig. 20(a). The condition of the contact that occurs in region EE' is similar, but moves from point E to E' on the flank of the gear towards the tip, Fig. 20(b). The maximum stress does not occur at point A or E, but rather in region $A'A$ or EE' .

5.3. Case 2: tooth contact under shaft misalignment

If two gears without any lead modification are paired on misaligned axes, the flank of the pinion which has a wider face-width will come into contact with the engaged flank of another gear on the edge of the face-width. The contact stress will consequently be concentrated on the edge. The distribution tendencies of the contact stress under different misalignment errors are summarized in Table 3. As expected, a concentrated contact stress occurs on the edge of the face-width which increases in value with an increase in the misalignment errors. The contact pattern changes from bone-shaped to delta-shaped, the area of contact is reduced, but the length of the contact pattern on the face-end edge is enlarged.

5.4. Case 3: gear pair with profile modification

In this case, the profile of the pinion is linearly relieved with a smoothing transition curve, both at the tip and the root of the pinion. The influence of the amount of modification on the contact characteristics of the gear pair is analyzed assuming ideal shaft alignment. The tip corner contact of the tooth pairs is also considered in the analysis.

5.4.1. Variation of load sharing during gear meshing

The variation in the shared loads the individual contact tooth pairs with different amounts of relief is shown in Fig. 21. The amount of relief affects the number of loaded tooth pairs in contact. A small amount of relief, e.g. 0.001 mm, retains the characteristics of that obtained without relief under a loaded contact ratio of about 2.12. With an increased relief amount, e.g., 0.005 mm, the loaded contact ratio is reduced to about 1.98, almost equal to 2, in other words, with almost no single tooth pair in loaded contact. The variation of the shared load in the region of double tooth-pair contact can be regarded as two different tendencies according to the conditions that one of the tooth pairs is in the relieved region or not. The shared load of a gear pair with a larger amount of relief (e.g., 0.009 mm) typically shows the variation tendency of a modified gear pair having a normal contact ratio (less than two). Not only the variation tendency in the region of double tooth pair contact (regions $A'B'$ and $D'E'$ in diagram), but also in the region of single tooth pair contact can be clearly identified (region $B'D'$). The variation of the shared load illustrates a different gradient around the relief starting points R_a and R_f from the others in the region of double tooth pair contact. The shared load in the relief region is smaller than the non-relief region.

5.4.2. Variation of contact stress during gear meshing

Fig. 22 shows the variation of the maximum contact stress with different amounts of relief of the tooth profile. There is a decrease in the concentrated contact stress due to tip corner contact before beginning of contact (A) and after end of contact (E) with a smaller amount of 0.001 mm, which also even disappears with larger amounts of relief with a corresponding decrease in the length of the line of action. The contact stress is reduced in the relief region and increased in the non-relief region with a larger relief amount.

The 3D diagrams of stress distribution on the flanks with different amounts of modification are illustrated in Table 4. The teeth in tooth pair 1 engage each other in the transition arc region. The corresponding contact stress along the tooth profile varies more asymmetrically with increased amounts of relief because of the asymmetric distribution of the separation distances among the flanks. The maximum stress still occurs on the input side of the shaft due to the influences of twist and stress concentration

on the end edge of the face-width. In contrast, with tooth pair 2 contact occurs at the non-relieved region. The contact stress distribution is similar to a saddle-shaped.

5.4.3. Transmission errors

There is variation in the loaded transmission errors (LTE) of the gear pairs with different amounts of relief during gear meshing, as the diagram shown in Fig. 23 and the values listed in Table 5. The tendency is similar to the variation of shared loads; please refer to Fig. 21. The peak-to-peak value of the LTE increases with a larger amount of relief, or in other words, with a reduction in the loaded contact ratio. This effect will be noticeable if the loaded contact ratio is less than 2. This means that an increased amount of relief will lessen the advantages of gear drives having a high contact ratio, such as the reduction of noise or vibration.

5.5. Case 4: lead-crowned gear pair under shaft misalignment

The contact characteristics of gear pairs with lead-crowning and misaligned shafts are analyzed. The non-modified tooth profile and the tip corner contact are also considered in this case.

5.5.1. Variation of load sharing during gear meshing

It can be recognized from the analysis results shown in Fig. 24 that the amount of crowning has no influence on the load sharing under the same amount of misalignment. In other words, the lead crowning does not affect the load sharing of the individual tooth pair, but only the positions of the contact points on the flanks, as well as the load distribution on the flank due to the stiffness of the shaft.

5.5.2. Variation of contact stress during gear meshing

The variation of maximum contact stress with different amounts of shaft misalignment shares the same tendency, as shown in Fig. 25. The location of the contact point is shifted towards the face-end with a larger amount of the shaft misalignment. As a consequence, there is a slight increase in the maximum stress of the contact ellipse, because of the increased curvature relation. The results corresponding to the diagrams for stress distribution on the flanks are listed in Table 6. With an increased misalignment amount, the contact pattern nearly remains, but its location moves towards the face-end.

The contact characteristics of the gear pairs with and without lead crowning in the presence of various values of shaft misalignment are compared in Fig. 26. The contact pattern of the non-modified gear pair changes from saddle-shaped to delta-shaped and tends to move towards the face-end with the increased value of misalignment. In the case of the misalignment 0.015 mm, the maximum contact stress occurs on the edge of face-end with the value of about 1061 N/mm², i.e., typical non-Hertzian contact (Table 3). This unfavorable contact can be improved by using lead crowning. The contact pattern of the modified gear pair remains ellipse-shaped and also moves towards the face-end. But the maximum contact stress is increased to the value of about 731 N/mm² (Table 6).

5.5.3. Transmission errors

The loaded transmission error influenced by the different crowning amounts and the misalignment deviations is compared with the results listed in Table 7. Both an increase in the amount of crowning and an increase in the misalignment deviation will enlarge the peak-to-peak value of the loaded transmission errors, but the difference is not large. On the other hand, the LTEs vary similarly to the diagram shown in Fig. 27, because the amount of deviation only slightly affects the meshing stiffness.

5.6. Case 5: gear pairs with lead and profile modification

In this case, a relief amount of 0.005 mm, a crowning amount of 0.015 mm and a shaft misalignment of 0.01 mm are selected for analysis of the contact characteristics. Figs. 28 and 29 show the results of the variation of the shared loads and the maximal contact stress, respectively. The results are similar to the case of the gear pair only with profile modification, compared with Figs. 21 and 22. It is clear to see that there is no rapid increase in the shared loads and no occurrence of contact stress at the beginning and end of tooth contact. The influence of the shaft misalignment on the contact stress is also compensated by lead crowning.

6. Conclusions

The proposed computerized LTCA approach has been successfully used to accurately simulate the contact characteristics of HCR spur gears. The model is based on the influence coefficient method and considers both the deformation of the teeth and the twist of the shaft. The results of the systematic analysis of different cases enable us to draw the following conclusions:

- (1) The results of the analysis show that the LTCA model is reliable, complete and time-saving for the calculation in comparison to the other mentioned LTCA approaches and the FEM. On one hand, the proposed LTCA approach considers the two dimensional condition of the flanks, while the other computerized LTCA approaches, e.g., LDP or LVR, focus only on flank condition along the major axis of contact ellipse. The contact patterns and also the two-dimensional stress distribution can be thus simulated, especially the non-Hertzian contact like the tip corner contact discussed in the paper. In this sense, the proposed approach is

also like FEM. On the other hand, the proposed approach can avoid the expensive works on establishing the finite element model with the accurate nodes to obtain better analysis results. It is very suitable to be used for efficient analysis of HCR gears with micro-level flank modification.

- (2) The HCR spur gears have better contact characteristics than normal-contact-ratio gears. Their contact behaviors are affected strongly by the number of contacting tooth pairs. Increasing the amount of relief can smooth the variation in shared loads and contact stresses during gear meshing, but the loaded contact ratio will be reduced. If the loaded contact ratio is below 2, the contact characteristics are the same as for normal spur gears. The contact stress will be reduced in the relief region and increased in the non-relief region with a larger relief amount. The peak-to-peak value of the loaded contact ratio will be also increased. It is suggested that the profile relief of HCR gears should be designed based on a loaded contact ratio not less than 2.
- (3) The influence of the shaft twist deformation on the contact characteristics of the spur gear pairs can be also identified by using the proposed LTCA approach. The distributed contact stress near the face-end where the torque is applied is enlarged. This effect is noticeably to identify in the case of line contact, but less so in the case of lead crowning.
- (4) The non-Hertzian contact patterns and the corresponding distributed stresses of spur gears without lead crowning and profile relief can be simulated by the proposed LTCA approach for cases of face-end edge contact in the presence of shaft misalignment and tip corner contact, respectively. The maximum concentrated stress due to tip corner contact does not occur at the beginning (position A) or end (position E) of contact on the normal line of action, but shortly before position A or after position E.
- (5) There is a slight increase in the peak-to-peak value of the loaded transmission errors with an increased amount of flank modification and/or shaft misalignment. The tendency of the variation during the gear meshing remains unchanged.

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